8.2 Intestinal Amoeba – *Entamoeba Histolytica*

8.2.1 Geographical Distribution

It is Worldwide in distribution they are more common in the tropics than elsewhere. It is found wherever sanitation is poor.

8.2.2 Habitat

Trophozoites of *E.histolytica* live in the mucous and submucous layers of the large intestine of human.

8.2.3 Morphology

E.histolytica occurs in 3 forms as Trophozoite, Precyst and Cyst.

Trophozoite: It is the growing or feeding stage of the parasite. It is the only form present in tissues. It has no fixed shape. They vary in size from 18 to 40µ, average being 20 to 30µ. The cytoplasm is usually described as outer ectoplasm and inner endoplasm (Figure 8.1). The endoplasm contains nucleus, food vacuoles, erythrocytes, occasionally leucocytes and tissue debris. The nucleus is characterised by evenly arranged chromatin on the nuclear membrane and the presence of a small, compact, centrally located karoyosome (It is a DNA containing body, situated peripherally or centrally within the nucleus). Trophozoites exhibits active crawling or gliding motility by forming finger-like projections called Pseudopodia.

The trophozoite reproduce by binary fission in every 8 hours. Trophozoites survives upto 5 hours at 37°C and are killed by heat, drying and chemical sterilization. Even if live trophozoites from freshly passed stools are ingested, they are rapidly destroyed in stomach and cannot initiate infection. Therefore, the infections is not usually transmitted by trophozoites.

Precyst

Trophozoites undergo encystment in the intestinal lumen. Encystment does not occur in the tissue or in feces outside the body. Precyst is smaller in size about 10 -20 µm in size. It is round or oval in shape. The endoplasm is free of red blood cells and other ingested food particles (Figure 8.1). The nuclear structure retains the characteristics of the trophozoite.

Cyst

Precyst secretes a highly refractive cyst wall around it and becomes a cyst. A mature cyst is a quadrinucleate spherical body. The cyst begins as a uninucleate body but soon divides by binary fission and develops into binucleate and quadrinucleate bodies (Figure 8.1). The cytoplasm of the cyst is clear and hyaline (translucent) and the nuclear structure retain the characteristic of the trophozoites.

The mature quadrinucleate cyst, passed in the stool, do not undergoes any further development and remain alive for several months in the soil or in environment where they were deposited. The mature quadrinucleate cysts are the infective forms of the parasite.

8.2.4 Life – Cycle of Entamoeba Histolytica

E. histolytica passes its life cycle only in one host, the human.

Infective form: Mature quadrinucleate cyst.

Mode of transmission: Ingestion of food and water contaminated with cyst.

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